

Financial cycle and business cycle: An empirical analysis based on the data from the U.S

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ABSTRACT

In this paper, we first study the relationship between the financial cycle and the business cycle in the time and frequency domain. Then we also explore the interactions and dynamic mechanisms of the financial cycle, the business cycle, real interest rate and exchange rate by the VAR model. The empirical results show that the financial cycle is closely related to the business cycle, especially at medium-term frequencies (8–30 years), the business cycle leads the financial cycle with a high positive correlation. However, the relationship between them is not significant during the Great Moderation at business-cycle (2–4 years). In addition, the financial cycle not only becomes a main driver of real interest rate, the financial cycle and the business cycle, but also serves as an important source of the business cycle fluctuations. In general, our results lay some theoretical foundation for the policy practice of financial and economic stability.

1. Introduction

The interplay between the financial sector and real economic activity has been extensively studied in the literature (Bernanke and Gertler, 1989; Bernanke et al., 1996; Kiyotaki and Moore, 1997). The studies show that wealth and substitution effects can be amplified in a world with financial friction because of changes in access to external financing. Changes in the supply of external financing can affect corporations and households, and thereby business cycles. In particular, after the global financial crisis of 2007–2008, the financial cycle and its interaction with the real economy have stimulated new interest. Reinhart and Rogoff (2009) review the performance of the real economy and financial variables during the financial crisis. From an empirical viewpoint, many studies concentrate on the intersection between credit and the business cycle. Helbling et al. (2011) describe one perspective on links between credit markets and global business cycles, namely that the credit shocks in the U.S. played an important role in shaping U.S. business cycles during the 1991 recession. Claessens et al. (2012) provide links between the business cycle (GDP) and financial cycles (credit, house prices, and equity prices) using an extensive database covering 44 countries for the period 1960:1–2010:4. The authors report that there are strong links between different phases of business cycle and financial cycle, recessions associated with financial disruptions tend to be longer and deeper than

other recessions, and that recoveries associated with rapid growth in credit and house prices are often stronger. These findings show the importance of financial market developments for the real economy. Borio (2014) concludes the main features of the financial cycle: (i) it is most parsimoniously described in terms of credit and property prices; (ii) it has a much lower frequency than the traditional business cycle (2–8 years); (iii) its peaks are closely associated with financial crises; and (iv) it helps to detect financial distress risks in real time.

However, it is still challenging to choose which financial variables and methods to use when constructing the financial cycle. From the existing literature, we can see that a single financial variable (credit, credit-to-GDP, house prices, equity prices, etc) is generally chosen to represent the financial cycle or construct synthetic financial cycle in many studies. There are three main methods of analyzing the characteristics of the financial cycle: (i) the analysis of turning points (Drehmann et al., 2012), (ii) frequency-based filter methods (Drehmann et al., 2012; Aikman et al., 2015), and (iii) spectral analysis (Strohsal et al., 2015). However, the limitations of these approaches are obvious. The turning point approach requires a pre-specified rule to be applied to an observed time series to find local maxima and minima. Frequency based filter methods require a pre-specified frequency range at which the financial cycle is assumed to operate (Strohsal et al., 2015). The spectral analysis approach has an advantage over frequency-based filter methods

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in that no a priori assumption is needed on the frequency range at which the financial cycle is assumed to operate. However, the Fourier transform only provides information about how strongly each frequency exists in the time series - it does not locate the points in time when these frequency components exist. The Fourier transform does not therefore provide any information regarding how the frequency content of the variable of interest evolves over time. Because of this, one has to split the data into sub-samples to analyze possible changes in the characteristics of the financial cycle over time (Verona, 2016).

In this paper we study the relationship between the financial cycle and the business cycle. First, we use macro-financial variables-private credit, private credit-to-GDP ratio, house prices, and share prices-to construct four synthetic financial cycles and select the best by comparing the performance in predicting economic recessions. We exploit wavelet analysis to analyze the duration of the financial cycle and how such duration has evolved over time. In addition, we use the wavelet coherence to analyze the relationship between the financial cycle and the business cycle in the time and frequency domain. Further, using the VAR model, we examine the interaction mechanism between the financial cycle, the business cycle, real interest rate, and exchange rate. In particular, since the financial cycle, real interest rate, and exchange rate are all representative financial variables that describe financial activities, this analysis will also help us deeply understand the finance system. Finally, based on the above analysis, we use the extended IS equation to analyze the impact of the financial cycle on the business cycle. Our paper thus contributes to a growing literature on the topic of the relationship between the financial cycle and the business cycle.

Rest of the paper is organized as follows. In section 2 we describe data collection. Section 3 explains construction and selection of the financial cycle proxies. Section 4 presents the methods used in this paper. In section 5 we report the results of the empirical analysis. Finally, section 6 concludes.

2. Data

First, we include two of the most common financial variables as proxies of the financial cycle: (i) credit in the private, non-financial sector from all sectors; (ii) residential property prices. The reasons for choosing these variables are as follows. First, the most parsimonious description of the financial cycle is in terms of credit and property prices (Borio, 2014). These variables tend to co-vary rather closely with each other, especially at low frequencies. At the same time, credit is an important variable that connects savings and investment, and can be used as a measure of the volatility of financial markets (Gorton and He, 2008). Real estate is used as collateral for credit, and property price fluctuations affect credit, and induce pro-cyclicality of credit and real estate prices. Second, SchularickTaylor (2011) point out that credit-to-GDP has an early warning for the crisis. At the same time, credit-to-GDP as an approximate measure of leverage in the macro-economy can be used as an indirect indicator of the absorptive capacity of the financial system. Finally, Drehmann et al. (2012) find that share prices do not fit as a proxy component of financial cycle because they exhibit a comparatively higher volatility at short term frequencies and co-move far less with the other two series, however, Schüler et al. (2015) argue that share prices do share important common cyclicalities with credit and residential prices. In addition, stock returns are often included in early warning exercises (Tólló et al., 2018). So we also include share prices as swings in stock prices are typically associated with boom-bust cycles in the financial markets (Voutilainen, 2017).

All series are in real terms (deflated by the GDP deflator) and in logs. The exception is the credit-to-GDP ratio, which is expressed in percentage points. Annual growth rates (year-over-year) of all series are obtained by taking four-quarter differences for each time series in logs. Real GDP growth (year-over-year) serves as the representative variable for the business cycle. We use quarterly data over the sample period 1970:1–2018:4, where credit (Credit to Private non-financial sector from All sectors) and credit-to-GDP (Credit to Private non-financial sector from

All sectors - Percentage of GDP) is obtained from BIS, house prices (residential property prices) and share prices are from OECD, and GDP deflator and real GDP are from FRED.

3. Construction and selection of financial cycle

We first divide the four common financial variables into four groups: (i) credit, house prices, and share prices; (ii) credit-to-GDP ratio, house prices and share prices; (iii) credit, credit-to-GDP ratio, house prices and share prices; (iv) credit, credit-to-GDP ratio and house prices. Then we use principal component analysis (PCA) to construct a composite financial cycle and denote by FC1, FC2, FC3 and FC4 respectively. Second, from the perspective of predicting economic recessions, we select the best synthetic financial cycle measure.

Since the amplitude of each financial time series is different, especially the fluctuation of share prices is significantly larger than other time series, the data can't be simply aggregated, then, the data is normalized by the standard min-max method:

$$VN_{i,t} = \frac{100(V_{i,t} - \text{Min}(V_i))}{\text{Max}(V_i) - \text{Min}(V_i)} \quad (1)$$

where Min (V_i) and Max (V_i) denote the minimum and the maximum, respectively, in the sample period; VN is the variable's normalized value. Equation (1) process converts all variables to the range of [0, 100] and an increase (higher value) means an improvement in financial conditions while a decrease means the opposite. The results of PCA are reported in Table 1, and the synthetic financial cycles are plotted in Fig. 1. On this basis, the financial cycles are further transformed into a gap value to more clearly show the financial cycles. Specifically, the HP filter is used to obtain the cycle series, and at the same time, in order to be consistent with the macro-prudential literature (Basel Committee on Banking Supervision, 2010; Borio, 2014), the smoothing parameter is 400,000, which preserves the medium-term component of the financial cycle. Schüler (2018) also shows that the Basel III specification of the HP filter induces, by using a smoothing parameter of 400,000, longer duration cycles that last up to 30 years.

In the following, unless otherwise specified, the financial cycle refers to the fluctuation term obtained by HP filter. Fig. 2 shows the financial cycles obtained by HP filter.

From Figs. 1–2, we can see that the financial cycles have higher correlation and similar behavior. It can be seen from Figs. 1–2, sometimes, the banking crisis did not occur after the peak of the financial cycle, however, the economic recessions occurred in 4 years after the financial cycle peaks. As we see, there was a peak reaching in 2005, a global financial crisis occurred two years later and led to a greater recession. So, we will choose the potentially best synthetic financial cycle by predicting recessions.

We take the recession dates from the National Bureau of Economic Research (NBER) and employ the probit model, which is characterized by the following equation:

$$\Pr(NBER_t = 1) = \Phi(C + FC_t) \quad (2)$$

where NBER is a binary variable, that is, recession occurs ($NBER = 1$) or it does not ($NBER = 0$), C is a constant term, FC is the synthetic financial cycle and Φ denotes the cumulative density function of the standard normal distribution. To determine the power of each financial cycle index in explaining economic recession events, we estimate the specification for each financial cycle.

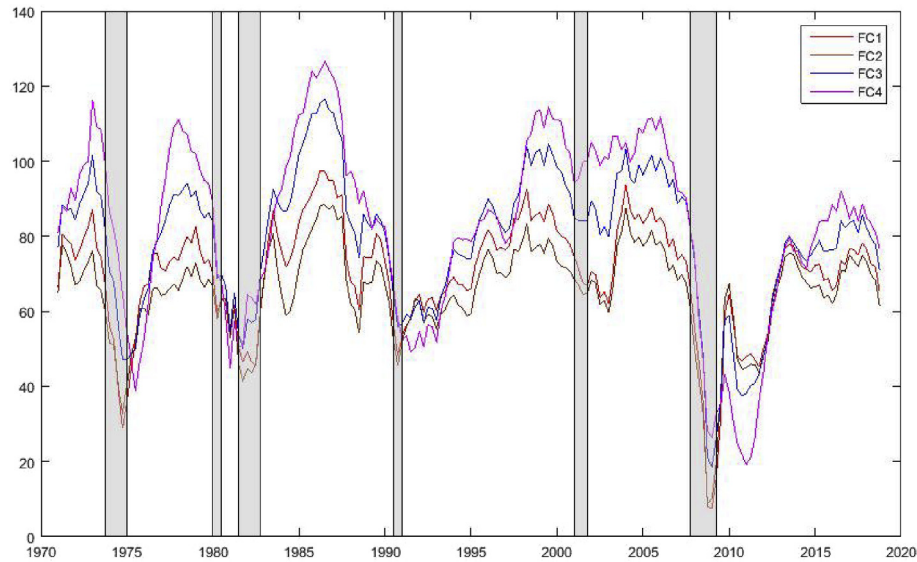
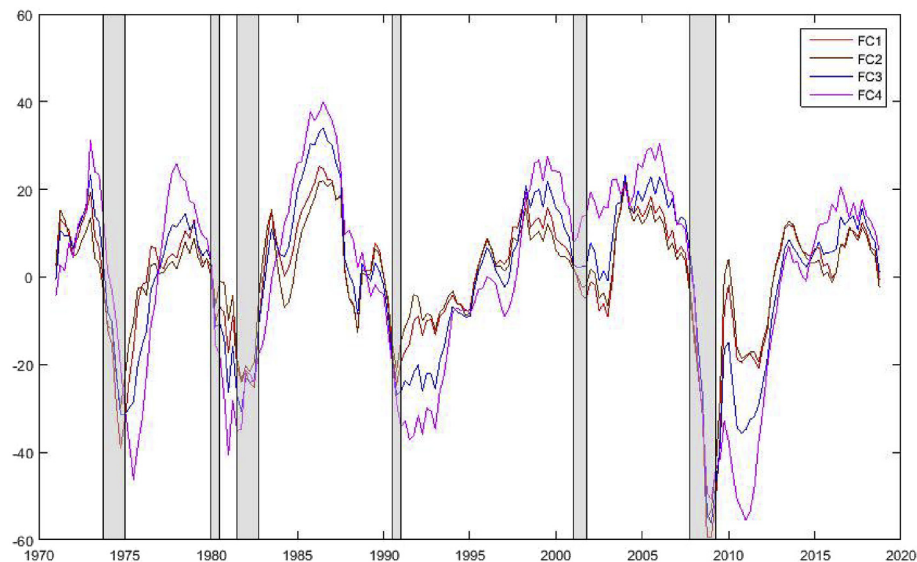
We consider 17 time horizons for the fact that recessions were occurred within 4 years after the peak of financial cycle, and specify for each synthetic financial cycle measures a special model, lagging the independent variables up to 16 quarters. To judge the model's adequacy and the goodness of fit, we determine "Area Under the Receiver Operating Characteristic" (AUROC) at the same time. The AUROC summary

Table 1

Principal component analysis for constructing the synthetic financial cycles.

	FC1		FC2		FC3		FC4	
	PC1	PC2	PC1	PC2	PC1	PC2	PC1	PC2
Eigen value	1.710	0.892	1.534	0.975	2.237	1.015	2.283	0.581
Proportion	0.570	0.297	0.511	0.325	0.582	0.254	0.761	0.194
Cumulative proportion	0.570	0.867	0.511	0.836	0.582	0.836	0.761	0.955

Note: The principal components (PC1 and PC2) are used to construct FC, capturing over 80% of the sample variation.

**Fig. 1.** Financial cycles and NBER Recessions (Shaded areas).**Fig. 2.** Financial cycles (HP with lambda 400,000) and NBER Recessions (Shaded areas).

statistics are bounded between 0 and 1, and higher AUROC values reflect more informative models. A value of 1 represents a perfect fit, however, a value below 0.5 shows an uninformative specification.

In Fig. 3, we plot the AUROC paths of the four synthetic financial cycle measures. All synthetic financial cycle measures are informative and tend to give reliable signals ahead of recessions indicated by AUROC values higher than 0.5 except at the one-year horizon for FC1 and FC2. In detail, for 0, 1, 2, 3, and 12–16 quarters horizons, the AUROC values of both FC1 and FC2 are higher than FC3 and FC4. The higher the AUROC

curve, the better is the performance of the model. Although no financial cycles perform better than others at each time horizon, on average, FC1 has an AUROC value of 0.720, while FC2, FC3 and FC4 get the AUROC values of 0.716, 0.710 and 0.683. In general, FC1 is in the top ranks on the average. Then we conclude that FC1 consisting of the real credit growth, real house prices and real share prices seems to be the best choice for the synthetic financial cycle.

In the following sections, FC1 is denoted by FC, which is described as the financial cycle in.

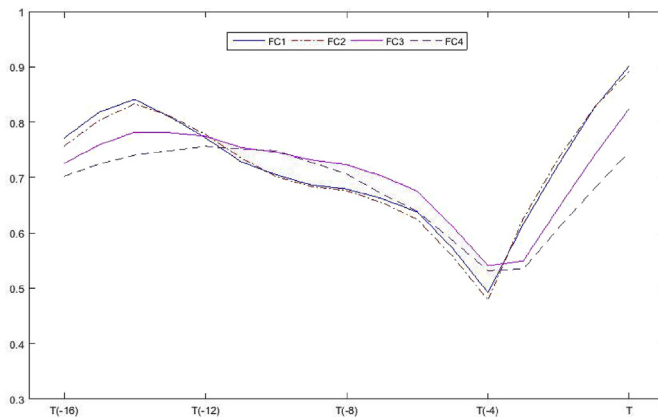


Fig. 3. Auroc of financial cycles.

4. Methods

In order to systematically analyze the relationship between the financial cycle and the business cycle, this paper adopts three methods: wavelets analysis, VAR model, and extended IS equation.

4.1. Wavelets analysis

Applications of the continuous wavelet transform (CWT) to economics have increased since the start of the 2000s (see Aguiar-Conraria et al., 2012 for more details). In this paper, we only describe the key points of wavelet analysis following the work Aguiar-Conraria and Soares (2014).

Given a time series $x(t)$, its CWT is defined by the following function:

$$W_x(\tau, s) = \int_{-\infty}^{+\infty} x(t) \frac{1}{\sqrt{|s|}} \psi\left(\frac{t-\tau}{s}\right) dt \quad (3)$$

where ψ is a mother wavelet, the bar denotes complex conjugation, s is a scaling that controls the width of the wavelet, and τ is the location parameter which describes where the wavelet is centered.

The specific mother wavelet we use in this paper is a complex-valued function selected from the Morlet wavelet family, and corresponds to the particular choice of $\omega_0 = 6$. See Aguiar-Conraria and Soares (2014) for a discussion of some desirable properties of this wavelet, which make its use attractive.

$$\psi_{\omega_0}(t) = \pi^{-\frac{1}{4}} e^{i\omega_0 t} e^{-\frac{t^2}{2}} \quad (4)$$

In analogy with the terminology used in the Fourier case, the (local) wavelet power spectrum is defined as

$$WPS_x(\tau, s) = |W_x(\tau, s)|^2 \quad (5)$$

This gives us a measure of the variance distribution of the time-series in the time-scale (frequency) plane.

The wavelet power spectrum can be average over time for comparison with classical spectral methods. The global wavelet power spectrum can be obtained by taking the averaging over all times:

$$GWPS_x(s) = \int_{-\infty}^{+\infty} |W_x(\tau, s)|^2 d\tau \quad (6)$$

To detecting and quantify relationships between two non-stationary time-series, we derive the concepts of cross-wavelet power, cross-wavelet coherency and wavelet phase-difference that enable us to deal with the time-frequency dependencies between two time-series.

For time-series $x(t)$ and $y(t)$, the cross-wavelet transform is defined as:

$$W_{xy}(\tau, s) = W_x(\tau, s) W_y(\tau, s) \quad (7)$$

where W_x and W_y are the wavelet transform of $x(t)$ and $y(t)$, respectively. $|W_{xy}|$ is defined as the cross-wavelet power, which depicts the local covariance between the time series at each time-frequency.

In analogy with the concept of coherency used in Fourier analysis, given two time series $x(t)$ and $y(t)$, we can define the complex wavelet coherency, and is denoted by $\rho(x, y)$, and

$$\rho(x, y) = \frac{S(W_{xy})}{|S(W_{xx})S(W_{yy})|^{\frac{1}{2}}} \quad (8)$$

where S denotes a smoothing operator in both time and frequency.

The absolute value of the complex wavelet coherency is called the wavelet coherency and is defined as follow:

$$R(x, y) = \frac{|S(W_{xy})|}{|S(W_{xx})S(W_{yy})|^{\frac{1}{2}}} \quad (9)$$

with $0 \leq R(x, y) \leq 1$.

With a complex-valued wavelet coherency, we can compute the phase-difference from the cross-wavelet transform by the following formulation:

$$\varphi_{xy} = \text{Arc tan} \left(\frac{I(W_{xy})}{R(W_{xy})} \right) \quad (10)$$

where $I(W_{xy})$ and $R(W_{xy})$ are the imaginary and real part of W_{xy} , $\varphi_{xy} \in [-\pi, \pi]$.

A phase-difference of zero indicates that the time-series move together at the specified frequency; if $\varphi_{xy} \in (0, \pi/2)$, then the time series are in phase (or positive) relation, but y leads x ; if $\varphi_{xy} \in (-\pi/2, 0)$, then x leads y ; if $\varphi_{xy} \in (\pi/2, \pi)$, then y leads x ; if $\varphi_{xy} \in (-\pi, -\pi/2)$, then x leads y ; if a phase-difference is π (or $-\pi$), that indicates an anti-phase (or negative) relation.

4.2. VAR model

The debate on the direction of causality between the financial cycle and the business cycle remains open. We estimate a VAR model, which not only has the advantage of providing evidence on the business cycle response to the financial cycle shocks and reactions in the financial cycle to the business cycle shocks, but also clarifies the causal relationship between the financial cycle and the business cycle.

We use the Akaike information criterion (AIC) and the Schwarz information criterion (SIC) to determine the lag in the VAR model, and find that the optimal lag period is 2. Fig. 4 shows that no root lies outside the unit circle, so VAR satisfies the stability condition.

4.3. Extended IS curve equation

Based on the VAR model, we follow Goodhart and Hofmann (2000) and Ma and Zhang (2016) in using the extended IS equation to determine the role of the financial cycle in aggregate demand in the following form:

$$y_t = \alpha_0 y_{t-1} + \alpha_1 FC_t + \alpha_2 ex_{t-1} + \alpha_3 (i_t - E_t \pi_{t+1}) + \varepsilon_t \quad (11)$$

where y_t , FC_t , ex_{t-1} , and $i_t - E_t \pi_{t+1}$ denote real GDP, financial cycle index, real effective exchange rate (narrow) with one lag period, and real interest rate respectively; π_t is the growth rate of GDP deflator; The term ε_t represents a demand shock. The coefficient α_0 measures the persistence of the business cycle, α_1 captures the impact of the financial cycle, α_2 captures the effect of changes in the real exchange rate (narrow), and α_3

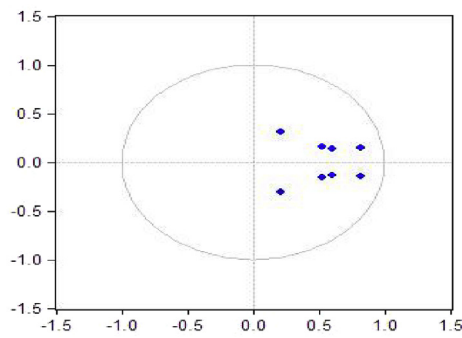


Fig. 4. Inverse roots of AR characteristic polynomial.

reflects the effect of the real interest rate on the business cycle. As has been found in the empirical literature (for example, [Borio, 2014](#)), the financial cycle exhibits co-movement with the business cycle. Therefore, the coefficient α_1 is expected to be positive and the coefficient α_3 is expected to be negative, because an increase in the real interest rate is usually associated with a contraction of the business cycle. We use the generalized method of moments (GMM) to estimate equation (11).

Compared with the traditional ordinary least squares method, the GMM method does not require complete knowledge of the underlying distribution of the data. It also controls for the endogeneity of explanatory variables, which helps to avoid bias due to model misspecification. Because all variables have periodic fluctuations after de-trending, the constant is not included in the regression equation. At the same time, instrument variables are the lag terms of financial variables. [Table 7](#) gives the estimation results from the GMM regression.

Table 2
Variance Decomposition of Financial cycle (FC).

Period	FC	BC	Real Interest Rate	EX
1	100.000	0.000	0.000	0.000
8	78.240	0.032	11.474	10.254
16	73.932	0.061	14.117	11.890
32	73.777	0.146	14.122	11.995

Table 3
Variance Decomposition of Business cycle (BC).

Period	FC	BC	Real Interest Rate	EX
1	8.638	91.362	0.000	0.000
8	33.119	44.168	20.546	2.367
16	31.026	40.610	25.834	2.530
32	31.101	40.451	25.657	2.791

Table 4
Variance decomposition of real interest rate.

Period	FC	BC	Real Interest Rate	EX
1	1.573	10.387	88.040	0.000
8	16.695	11.952	67.012	4.341
16	18.554	13.406	61.461	5.579
32	18.553	13.418	61.423	6.616

Table 5
Variance decomposition of exchange rate (EX).

Period	FC	BC	Real Interest Rate	EX
1	7.222	0.470	1.182	91.126
8	5.798	7.496	6.253	80.453
16	5.828	7.812	7.209	79.151
32	5.834	7.812	7.213	79.141

Table 6
VAR Granger causality/Block Exogeneity Wald Tests.

Dependent variable: FC			
Excluded	Chi-sq	df	Prob
BC	0.439	2	0.803
Real interest rate	11.327	2	0.004
EX	10.085	2	0.007
ALL	17.370	6	0.008
Dependent variable: BC			
Excluded	Chi-sq	df	Prob
FC	26.556	2	0.000
Real interest rate	15.135	2	0.001
EX	0.168	2	0.919
ALL	47.368	6	0.000
Dependent variable: Real interest rate			
Excluded	Chi-sq	df	Prob
FC	8.488	2	0.014
BC	2.032	2	0.362
EX	1.464	2	0.481
ALL	13.703	6	0.033
Dependent variable: EX			
Excluded	Chi-sq	df	Prob
FC	8.864	2	0.012
BC	3.665	2	0.160
Real interest rate	4.946	2	0.084
ALL	12.105	6	0.060

Table 7
The estimation results of the extended IS curve equation by GMM.

Variable	Coe_cient (t-Statistic)	Adjusted R2	J-statistic (p-value)
y (-1)	0.759*** (14.533)	0.761	0.042 (0.837)
FC	0.039*** (5.305)		
ex (-1)	0.047** (2.141)		
real interest rate	-0.214*** (-3.315)		

Note: Significance at 0:1, 0:05 and 0:01 levels are reported by *, ** and ***, respectively.

5. Empirical results

5.1. Results from wavelet analysis

Wavelet analysis allows one to take into account both frequency and time variations in a time series. We use continuous wavelet tools to analyze how the financial cycle has evolved over times, and the lead-lag relationship between the financial and business cycles. In [Fig. 5](#), which shows the wavelet power spectrum (WPS), time is on the horizontal axis and period cycles (in years) are on the vertical axis. Hotter colors (yellow and red) correspond to higher volatility and colder colors (green and blue) to lower volatility. The black contours mark significance at the 5 percent level, and the black dashed lines denote the cone of influence, which indicates regions influenced by edge effects where caution should be applied to interpretations. The white lines show the maxima of the undulations in the WPS, and provide an estimate of the cycle periods ([Claessens et al., 2012](#); [Verona, 2016](#)). [Figs. 5–6](#) present the results of the wavelets analysis.

The financial cycle exhibits three main cycles (middle plot in [Fig. 5](#)): one with a period of about 7 years between 1972 and 1982; a longer cycle starting in about 1985 with a period of 8–10 years, which is confirmed that the length of the financial cycle has been longer since 1985 for the economic globalization and financial liberalization; and a third permanent cycle with a period of about 16 years throughout most of the sample. At the same time, the GWPS (bottom plot in [Fig. 4](#)) confirms that the majority of the variability of the financial cycle has cycles with longer periods (more than 8 years).

[Fig. 6](#) shows the relationship between the financial cycle and the business cycle in the time and frequency domain. We group the frequency bands into three categories, that is ‘short term’ (4–8 quarters), ‘business cycle’ (2–4 years and 4–8 years), and ‘medium term’ (8–30 years). The short term category captures high frequency fluctuations under two

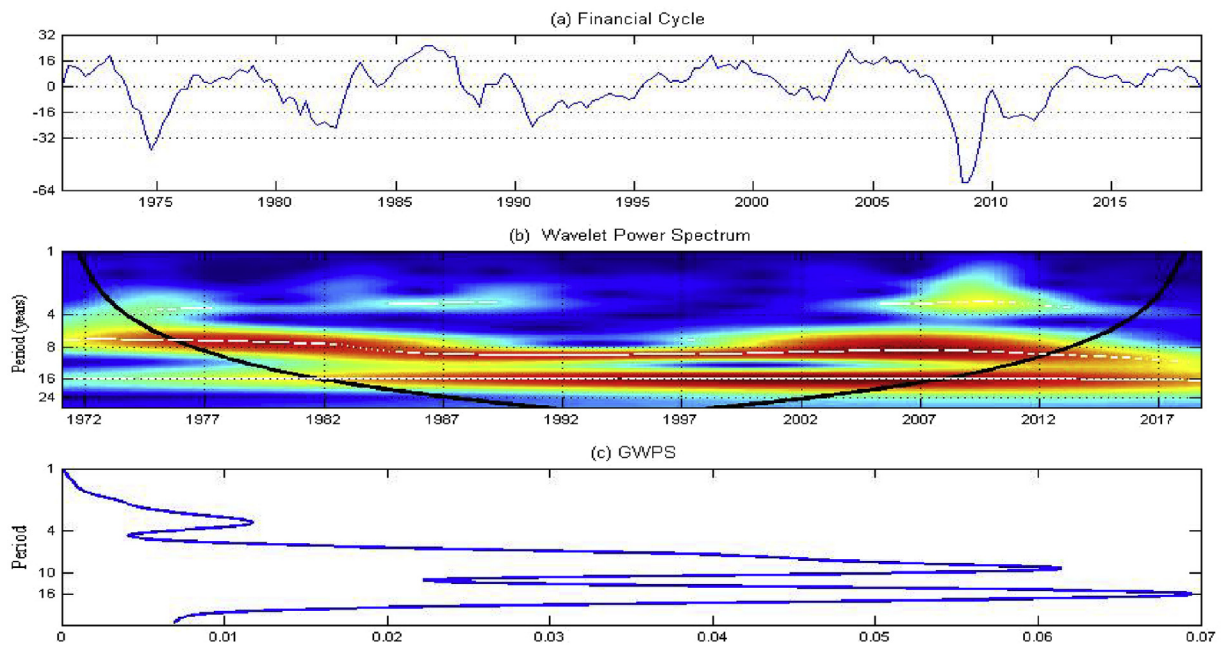


Fig. 5. Financial cycle, WPS and GWPS.

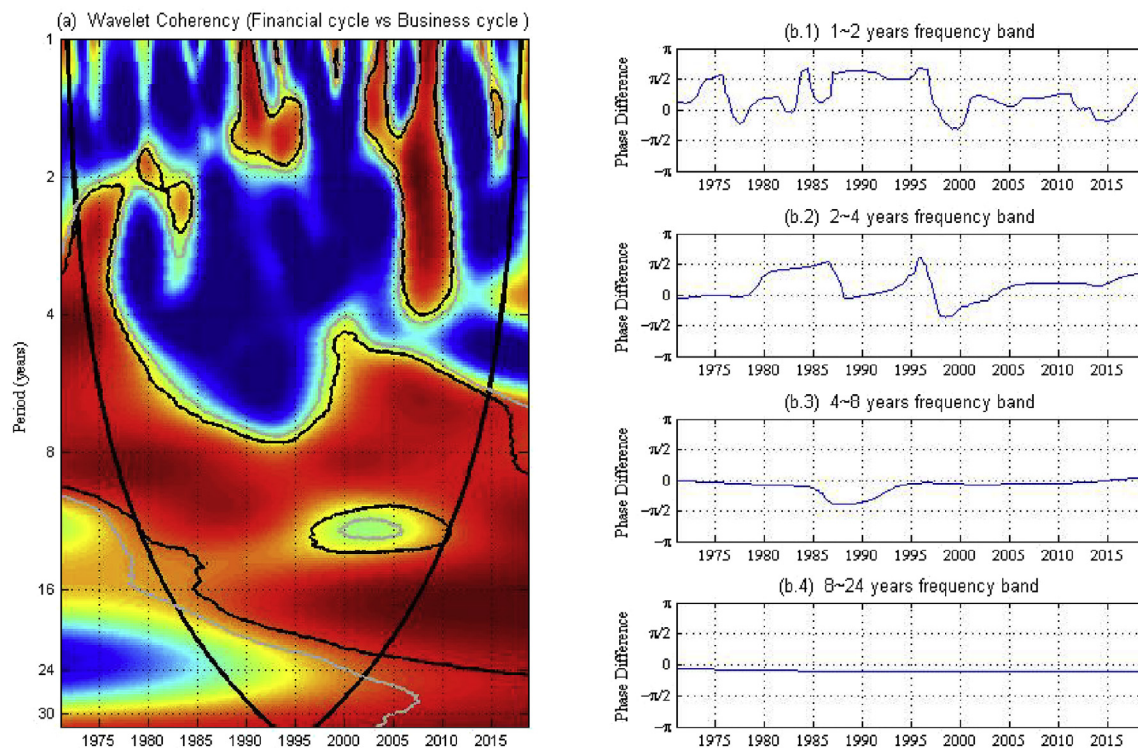


Fig. 6. The lead and lag relationship between the financial cycle and the business cycle.

years. The business cycle category covers fluctuations at frequencies between 2 and 8 years, which most macroeconomic research on the sources of aggregate movements focuses on. The medium-term category obtains fluctuations at frequencies between 8 and 30 years, which is the length of the financial cycle.

In short term, there are four small statistically significant regions in which the financial cycle and the business cycle (real GDP) present strong relationships during 1979–1980, 1989–1995, 2003–2005 and 2005–2010 respectively. It is worth noting that during the above four time periods, three economic recessions and two financial crises

occurred, including the savings and loan crisis and the global financial crisis. In terms of phase-difference, in short term frequencies, during 1979–1980, 2003–2005 and 2005–2010, the financial cycle leads the business cycle and has a positive correlation, however, it is the opposite between 1989 and 1995, the business cycle leads the financial cycle with a negative correlation.

At business-cycle frequencies (2–8 years), we further divided it into 2–4 years and 4–8 years according to the actual situation. In the frequency bands of 2–4 years, there are three main regions of significant coherency during 1971–1978, 1981–1984, and 2005–2010 respectively.

During 1971–1978, the financial cycle and the business cycle are almost synchronously, while in the other three regions, the financial cycle is leading the business cycle. It is worth noting that the relationship between the financial cycle and the business cycle is not significant from the mid-1980s to 2005. The relationship become significant again until 2006, after that, the Great Recession occurred during 2007–2009. However, the relationship disappeared again since 2011.

At business-cycle frequencies (4–8 years), the region of high coherency between the financial cycle and the business cycle occurs in most of the sample period. It is visible a gradual shift from the 4 year frequency band to the 8 year frequency band, which is reverted after 1994. We also observe a rather stable phase relationship at business-cycle frequencies (4–8 years): for most of the time, the phase difference is less than and close to 0, indicating that the business cycle slightly leads the financial cycle and that an expansion in the business cycle is associated with an expansion in the financial cycle. In fact, we can assume that the financial cycle and the business cycle show the behavior of co-movement almost at the 4–8 years frequency band.

At medium-term frequencies (8–30 years), the financial cycle has a strong relationship with the business cycle almost during the sample period, there is a large and significant region of coherency (except a small region between 1996 and 2010 at the 10–13 years frequency band), and the frequency band gradually increases from about 9 years to 30 years. At the same time, in the term of the phase difference, the phase relationship is very stable with the business cycle leading the financial cycle about 1–4 quarters.

From the above analysis, we know that there is a close relationship between the financial cycle and the business cycle at the 4–30 years frequency band, and the business cycle leads the financial cycle with a positive correlation. But in short-term frequencies (1–2 years), the relationship was not significant except during or after the economic downturn or financial crisis. At business cycle frequencies (2–4 years), the relationship between the financial cycle and the business cycle was not significant during the Great Moderation.

However, the positive (negative) relationship is not causal relationship. To further study the causality between the financial cycle and the business cycle, we use a VAR model to investigate the relationship.

5.2. Results from the VAR model

In this subsection, we first apply the VAR model to describe the dynamic mechanism among the financial cycle, the business cycle, real interest rate and exchange rate. Subsequently, we investigate the Grange-causality.

To analyze the dynamic mechanism among these variables, Fig. 7 shows the impulse response functions by using the Cholesky decomposition.

In first column in Fig. 7, we plot the impulse response functions for a positive shock in the financial cycle. For the shock of the financial cycle, the financial cycle gradually weakens after reaching its peak in the second quarter, and turns negative after 12 quarters, but the negative response is weak. The impulses of the business cycle and the real interest

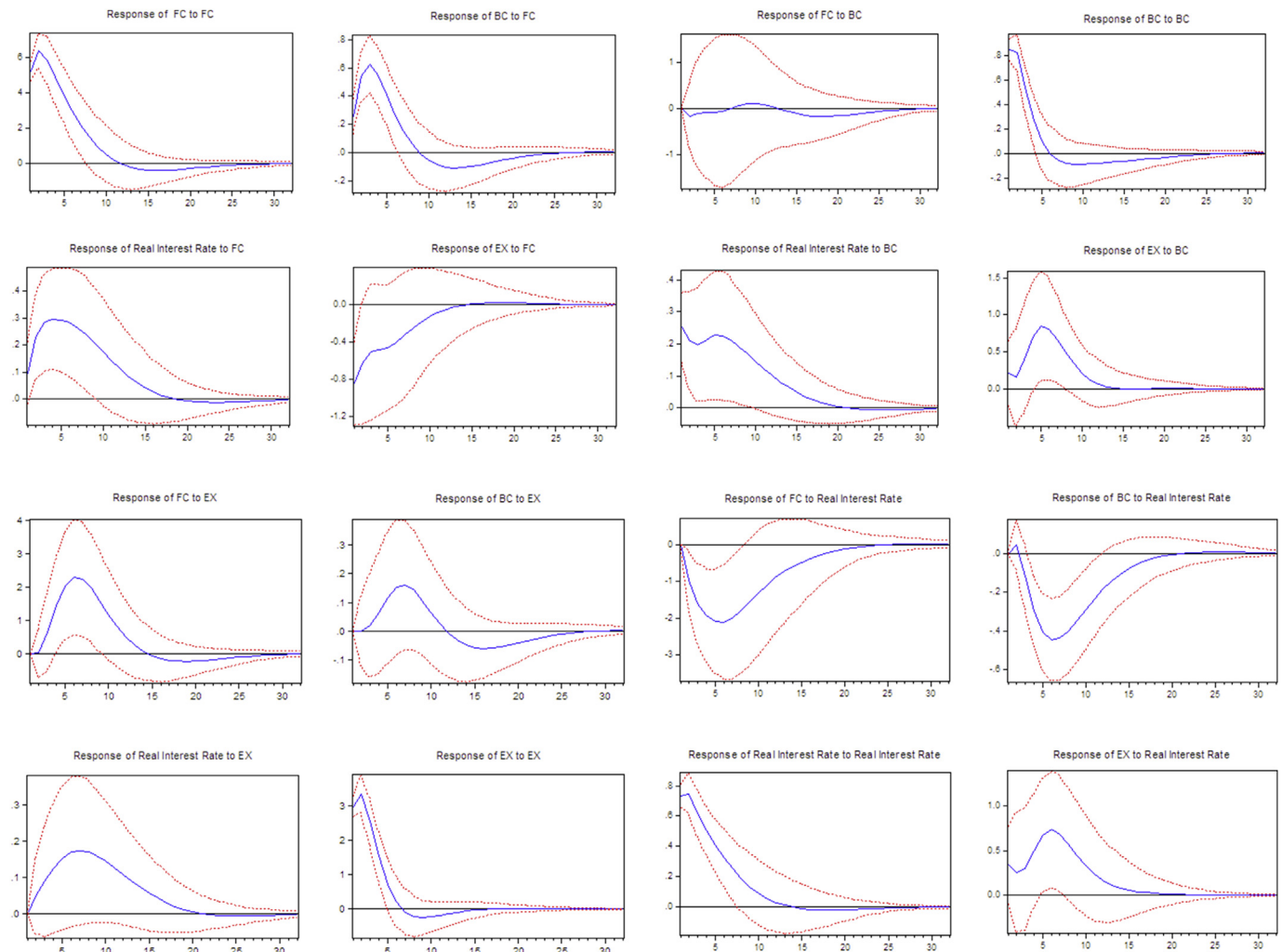


Fig. 7. Impulse-response functions. Note: FC, BC, and EX are for Financial cycle, Business cycle, and Exchange rate, respectively.

rate are positive for the first 8 and 18 periods, and the positive response time of the real interest rate is longer than the business cycle. The response of the exchange rate is negative and tends to zero after 13 quarters.

In the second column in Fig. 7, the impulse response function for a positive shock in the business cycle is shown. This clearly shows that the response of the business cycle tends to zero after 10 quarters. The response of the financial cycle to the business cycle is weak. This suggests that the interaction between the financial cycle and the business cycle is asymmetric, and the response of the business cycle to the financial cycle is strong. Both the interest rate and the exchange rate response positively to the business cycle, but the response of the interest rate lasts longer than that of the exchange rate; After reaching the maximum in the first quarter, and then to zero after 20 quarters. The exchange rate response reached its maximum after the fifth quarter, and then gradually decreases to zero.

In the third column in Fig. 7, one standard deviation of the real interest rate shock is shown. Both the financial cycle and the business cycle react negatively to the positive shock of the real interest rate, and reach the largest negative response in sixth and eighth quarter respectively, and then gradually turn to zero. The impulse of the exchange rate is reversed, reaching a positive peak in the sixth quarter. This indicates that there is a trade-off between the financial cycle and the real interest rate, and that there is also a trade-off between the business cycle and the real interest rate. These results are consistent with the economic theory.

Finally, for a positive shock of the exchange rate, the impulses of the financial cycle, the business cycle, and the real interest rate peak after 6–7 quarters, and then gradually weaken, wherein the shock response of the financial cycle and the business cycle turns negative after 15 and 12 quarters respectively. The exchange rate itself gradually weakens after reaching its maximum value in the second quarter, and tends to zero after 15 quarters.

Based on our analysis of the impulse response, we further analyze the degree of contribution for different periods by variance decomposition. The variance decomposition for each variable is shown in Tables 2–5.

In Tables 2–5, at the 32-quarters horizons, the contribution rate of the financial cycle to its own variance is gradually reduces from 100% and finally stabilizes at 73.8%, the power to explain the variance of the business cycle exceeds 31% after eighth quarter, and the contribution rate to the real interest rate and the exchange rate finally stabilizes at 18.6% and 5.8%, respectively. The contribution rate of the business cycle to the variance of the financial cycle is very small, no more than 0.15%, and the contribution rate of its own variance gradually decreases from 100% and finally stabilizes at around 40.5%. The explanatory power.

Of the variance of the real interest rate fluctuates less, it is stable at 25.7% in the ability to explain the variance of the business cycle and at around 7.2% from the 10th quarter in the ability to explain the variance of the exchange rate. The contribution rate of the real interest rate to the financial cycle and the business cycle is larger than that of the exchange rate. In particular, the contribution rate of the variance of the business cycle reaches more than 20% after 8 quarters, and the contribution rate of the exchange rate to the business cycle does not exceed 2.8%. Finally, the contribution rate of the financial cycle, the business cycle and real interest rate to the variance of the exchange rate is small, about 5.8%, 7.8% and 7.1% respectively.

In general, the financial cycle has a strong ability to explain the variance of the business cycle. In particular, the contribution rate to the variance of the business cycle rises rapidly from the beginning of 8.6% and stabilizes at 31.4% with a peak of 37.4%. Its own variance contribution rate decreases by 100% and stabilizes at 73.8%. In comparison, the business cycle shock has a lower contribution rate to other variables, and only accounts for 13.4% and 7.2% of the real interest rate and the exchange rate fluctuations in the long run. Furthermore, the business cycle's explanatory power for the financial cycle fluctuations is no more than 0.15%, which can be negligible.

Overall, the results in Tables 2–5 show that the contribution rate of

the financial cycle shock to the business cycle variance exceeds the real interest rate and the exchange rate, and the contribution rate to the real exchange rate variance exceeds the business cycle and exchange rate. In short, for the shock of each variable (except the exchange rate), except for its own influence, the impact of the financial cycle shock is the most significant. In addition, the volatility of the financial cycle is mainly explained by its own impact, and the contribution of other variables is less. These results indicate that the financial cycle shock has become a key driving force affecting the dynamics of the VAR model.

To further understand the transmission mechanism between the financial cycle, the business cycle, real interest rate, and exchange rate, Table 6 gives the results of Granger.

-causality test for the above four-variable VAR model. The results show that there is a two-way positive Granger causality between real interest rate and the financial cycle, and exchange rate and the financial cycle, but the real interest rate one-way positively affects the business cycle and the exchange rate, there is no significant cause between exchange rate and the business cycle.

The transmission mechanism revealed by the above-mentioned Granger causality test shows that the financial cycle plays a kernel and key role.

6. Conclusion

Based on U.S. quarterly data from 1970:1 to 2018:4, this paper divides four representative financial variables into four groups and uses principal component analysis to synthesize them into four composite indexes. The composite index with the best performance in predicting economic recession is selected as the financial cycle. On this basis, the relationship between the financial cycle and the business cycle is analyzed in detail. At the same time, we also studied the interaction mechanism of the financial cycle, the business cycle, real interest rate, and exchange rate. From empirical analysis, we obtain the following main conclusions:

First, the financial cycle index constructed in this paper has good predictive ability for economic recession within 0–16 quarters and is closely related to the business cycle, especially, at medium-term frequencies (8–30 years), the business cycle is leading the financial cycle and has a high correlation with the financial cycle.

Second, compared with real interest rate and exchange rate, the financial cycle plays an important role in financial and economic activities. Specifically, the fluctuation of the financial cycle not only becomes a key driver of the business cycles and real interest rate, but also the variance changes have a clearer and stronger explanatory power for the business cycles and the real interest rate. At the same time, it is worth noting that the ability of the business cycle to interpret the variance of the financial cycle is very weak, and there is a clear asymmetric behavior between them.

Third, compared with other variables (such as interest rates and exchange rates), the volatility of the financial cycle has a stronger explanatory power for the business cycle volatility. In the long run, about 31% of the volatility of the business cycle can be explained by the shock of the financial cycle, which shows that the shock from the financial system has become an important source of the business cycle fluctuations.

Overall, the empirical analysis in this paper expands the existing literature and further confirms the close relationship between the financial cycle and the business cycle, as well as the important impact of the financial cycle shock on macroeconomic fluctuations. These results also suggest that the financial cycle may be another indicator that could be useful to policymakers.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.econmod.2020.01.018>.

References

- Aguiar-Conraria, L., Soares, M.J., 2014. The continuous wavelet transform: moving beyond uni-and bivariate analysis. *J. Econ. Surv.* 28, 344–375.
- Aguiar-Conraria, L., Martins, M.M., Soares, M.J., 2012. The yield curve and the macro-economy across time and frequencies. *J. Econ. Dynam. Contr.* 36 (12), 1950–1970.
- Aikman, D., Haldane, A., Nelson, B., 2015. Curbing the credit cycle. *Econ. J.* 125, 1072–1109.
- Basel Committee on Banking Supervision, 2010. Guidance for National Authorities Operating the Countercyclical Capital Buffer. Bank for International Settlements.
- Bernanke, B., Gertler, M., 1989. Agency costs, net worth, and business fluctuations. *Am. Econ. Rev.* 79 (1), 14–31.
- Bernanke, B., Gertler, M., Gilchrist, S., 1996. The financial accelerator and the flight to quality. *Rev. Econ. Stat.* 78 (1), 1–15.
- Borio, C., 2014. The financial cycle and macroeconomics: what have we learn? *J. Bank. Finance* 182–198.
- Claessens, S., Kose, M., Terrones, M., 2012. How do business and financial cycles interact? *J. Int. Econ.* 87 (1), 178–190.
- Drehmann, M., Borio, C., Tsatsaronis, K., 2012. Characterizing the Financial Cycle: Don't Lose Sight of the Medium Term!, BIS Working Papers. June, p. 380.
- Goodhart C., Hofmann B., 2000. Financial variables and the conduct of monetary policy. Sveriges Riksbank Working Paper No.112.
- Gorton, G.B., He, P., 2008. Bank credit cycles. *Rev. Econ. Stud.* 75 (4), 1181–1214.
- Helbling, Thomas, Raju, Huidrom, M., Kose, Ayhan, Otrok, Christopher, 2011. Do credit shocks matter? A global perspective. *Eur. Econ. Rev.* 55, 340–353.
- Kiyotaki, N., Moore, J., 1997. Credit cycles. *J. Polit. Econ.* 105 (2), 211C248.
- Ma, Y., Zhang, J., 2016. Financial cycle, Business cycle and monetary policy: Evidence from four major economies. *Int. J. Finance Econ.* 21, 502–527.
- Reinhart, C., Rogoff, K., 2009. *This Time Is Different: Eight Centuries of Financial Folly*. Princeton University Press.
- Schularick, M., Taylor, A.M., 2011. 2011. Credit Booms Gone Bust Monetary Policy Leverage Cycles and Financial Crises, 1870–2008. NBER.
- Schüler, Y., Hiebert, P., Peltonen, T. 2015. Characterizing the financial cycle: multivariate and time-varying approach. In: ECB Working Paper Series, no. 1846
- Schüler, Y., 2018. Detrending and Financial Cycle Facts across G7 Countries: Mind a Spurious Medium Term! ECB, p. 2138.
- Strohsal, T., Proano, C.R., Wolters, J., 2015. Characterizing the Financial Cycle: Evidence from a Frequency Domain Analysis. Deutsche Bundesbank. Discussion Papers 22/2015.
- Tölö, Eero, Laakkonen, Helinä, Kalatie, Simo, 2018. Evaluating indicators for use in setting the countercyclical capital buffer. *International Journal of Central Banking* 14 (2), 51–112.
- Verona, Fabio, 2016. Time-frequency characterization of the U.S. Financial cycle. *Econ. Lett.* 144 (C), 75–79.
- Voutilainen, Ville, 2017. 2017. Wavelet Decomposition of the Financial Cycle: an Early Warning System for Financial Tsunamis. Bank of Finland Research Discussion Paper.